CAPSTONE PROJECT REPORT

Doctors don't write... They encrypt!

Digitally decrypting handwritten pharmaceutical scripts using deep learning.

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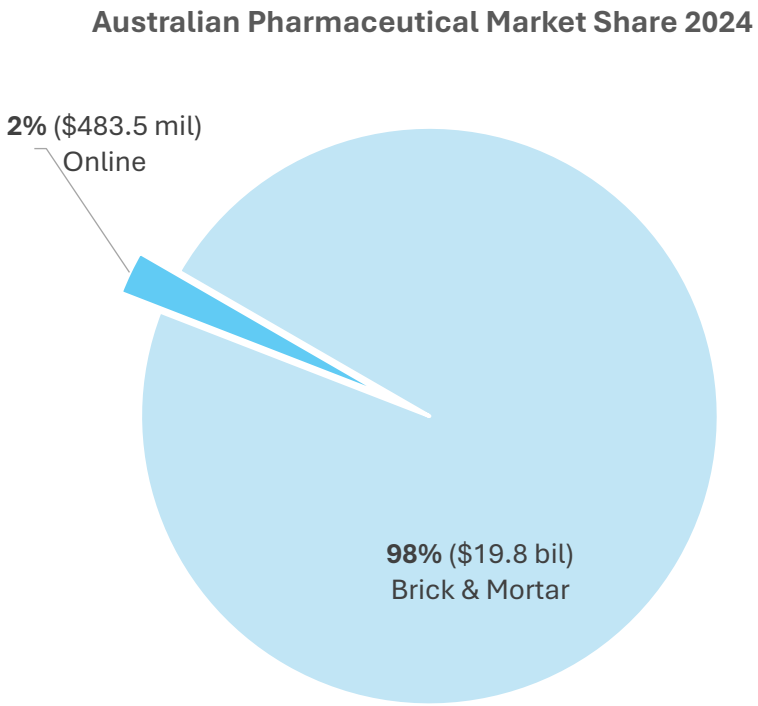
Problem Statement

Currently, online medication purchases in Australia rely on one of two methods: mailing a handwritten prescription or submitting an electronic prescription (eScript) issued by a doctor. eScripts were introduced during the COVID-19 pandemic to enable contactless prescribing and have seen rapid adoption. In 2024, approximately 189 million eScripts were issued, accounting for around 54% of all prescriptions nationwide.

Despite this growth, a significant proportion of medical practitioners continue to issue handwritten prescriptions. This creates a barrier to wider adoption of online pharmacy services, as patients using handwritten prescriptions are required to mail physical scripts, introducing delays and logistical friction.



The Australian pharmaceutical market was valued at approximately $19.3 billion in 2024, with only $483.5 million attributed to online pharmaceutical retail. This indicates that nearly 98% of transactions still occur through brick-and-mortar pharmacies, highlighting a substantial opportunity for digital expansion.

This project proposes the development of a handwriting recognition system that enables patients to submit handwritten prescriptions digitally via file upload or smartphone image capture. The system uses a deep learning convolutional neural network (CNN) to process prescription images and classify handwritten medication names against a

pretrained dataset. By automating the interpretation of handwritten prescriptions, the solution aims to remove the need for postal submission, digitally validate prescriptions, and allow patients to complete purchases immediately, resulting in faster dispensing and delivery of medications.

Industry / Domain

Within the online retail (e-commerce) landscape, the pharmaceutical and medical industries have been slower to adopt digital sales channels compared to other sectors. This is largely due to the highly regulated nature of the industry and the strict compliance requirements governing the sale and dispensing of prescription medications.

As outlined in the problem statement, online pharmaceutical retailers must adhere to rigorous guidelines for verifying prescriptions prior to dispensing medications. Any digital solution operating in this space must therefore meet regulatory standards for accuracy, security, and validation.

As part of this project, a prototype handwriting recognition model would be developed and demonstrated to relevant regulatory bodies to assess its suitability for real-world use. Regulatory approval would be a prerequisite for commercial deployment, enabling pharmaceutical retailers to integrate the tool into existing or planned e-commerce platforms while maintaining compliance with industry regulations.

Stakeholders

This project involves three primary stakeholder groups:

Government Regulators
These stakeholders are responsible for oversight, compliance, and approval of any system involved in prescription handling and medication dispensing:

- Therapeutic Goods Administration (TGA)
- Medicare
- Pharmaceutical Benefits Scheme (PBS)
- Australian Department of Health
- State and territory health departments

Clients / Customers
These stakeholders would be the commercial users of the solution:

- Pharmaceutical retailers, both with and without existing e-commerce sales channels

- E-commerce merchants seeking to expand into pharmaceutical products as a new category

End Users
These stakeholders are the primary users of the system:

- Patients submitting handwritten prescriptions digitally via file upload or smartphone image capture

Business Question

The core business question for this project is whether a new digital method can be introduced to allow patients to submit handwritten prescriptions online.

By offering handwritten prescription submission as a third option alongside mailed scripts and eScripts, online pharmacies may be able to increase customer adoption, reduce purchase friction, and capture a larger share of transactions currently handled by brick-and-mortar pharmacies.

A secondary business consideration is operational efficiency. This project also seeks to assess the extent to which manual prescription review and auditing can be reduced based on the accuracy and reliability of the handwriting recognition model, potentially lowering labour costs while maintaining regulatory compliance.

Data Question

The primary data question for this project is how accurately a model can classify handwritten medication names, with a specific focus on optimising **Top-1 accuracy**. Top-1 accuracy is treated as the primary performance metric because the highest-probability prediction must align with the medication name represented in the handwritten prescription. Maximising Top-1 accuracy increases confidence that the model's first output is correct, which is critical for automated prescription processing.

For this project, a **Top-1 accuracy threshold of approximately 85–90%** is considered a practical benchmark for enabling semi-automated approval, where high-confidence predictions can be accepted automatically and lower-confidence cases are flagged for human review.

To support this primary objective, additional evaluation metrics are used. **Macro F1-score** is employed to ensure balanced performance across all medication classes, preventing the model from favouring more frequent or visually distinct names. **Top-5 accuracy** is used to measure the likelihood that the correct medication name appears within the top five predictions when Top-1 classification fails. A high Top-5 accuracy (e.g. ≥ 95%) ensures that, in cases requiring manual review, the correct medication is highly

likely to be present among a small set of candidate predictions, significantly reducing the time required for human evaluation.

Data

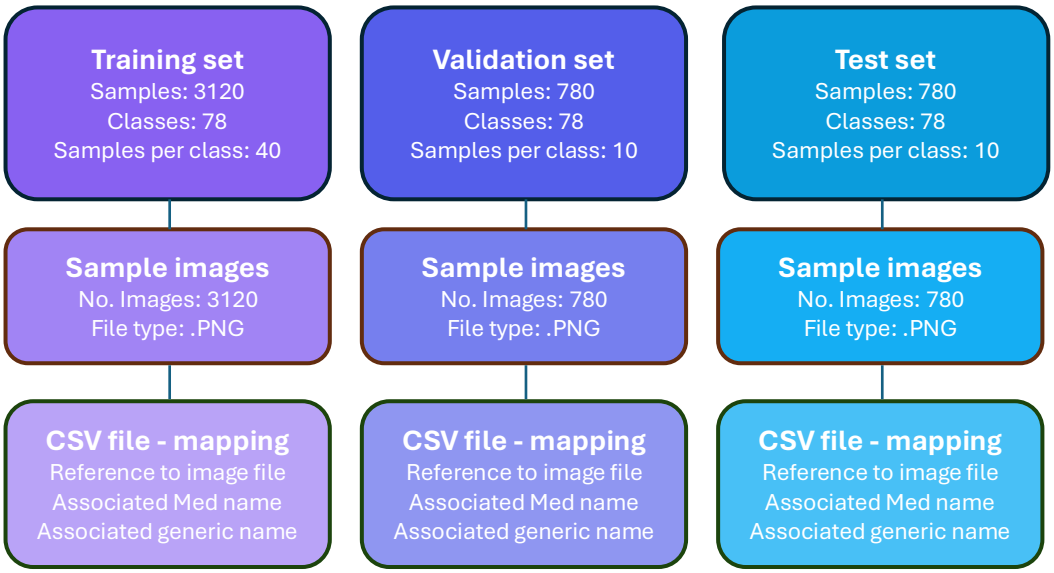
The dataset used for this project was sourced from Kaggle and selected to support an initial proof of concept, given the lack of accessible real-world prescription data within the project timeline. Preliminary visual inspection indicated sufficient variation in handwriting styles to provide a meaningful challenge for the model.

The dataset originates from a published study titled *"A Deep Neural Network Approach with Pioneering Local Dataset to Recognize Doctor's Handwritten Prescription in Bangladesh,"* which similarly applied a convolutional neural network (CNN) for handwritten prescription classification.

The dataset consists of scanned handwritten prescription images accompanied by a CSV file mapping each image filename to both a medication name and a generic name. For this project, the **medication name** was used as the target variable for model training. All image–label mappings were manually annotated and independently verified by multiple contributors to ensure labelling accuracy.

The chart below illustrates the structure of the dataset.

Doctors Handwritten Dataset Structure



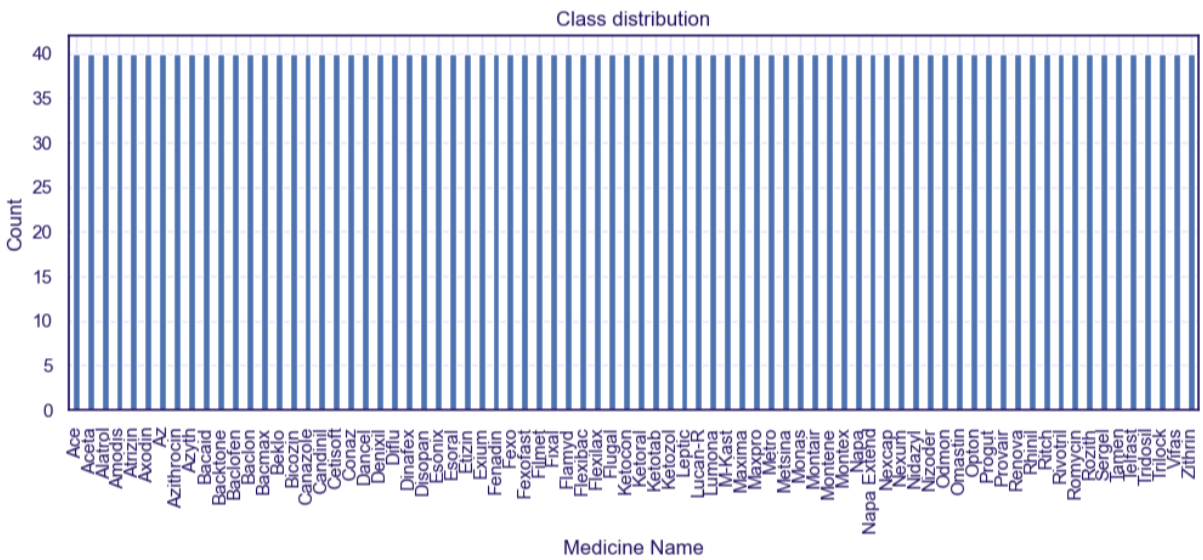
Exploratory Data Analysis & Data Process

The full dataset consists of 4,680 samples, split into three subsets: training, validation, and testing. The training set contains 3,120 samples, while both the validation and test sets contain 780 samples each.

Data set & Class Size

The dataset includes **78 distinct medication name classes**, with an even distribution of samples across the training, validation, and test sets. This balanced class structure helps reduce bias during training and supports fair performance evaluation across all medication names.

An example of the training set class distribution is shown below.



Additionally, a number of medication names exhibit similar prefixes, suffixes, and visual word shapes, which increases inter-class similarity.

Each dataset split contains:

- Image files in **.png** format
- A corresponding **CSV file** with three columns:
    - Image filename
    - Medication name (class label)
    - Generic medication name

For this project, the **medication name** label is used as the prediction target.

TRAIN SHAPE

(3120, 3)

TEST SHAPE

(780, 3)

VAL SHAPE

(780, 3)

Missing Values and Duplications

Data quality checks were performed to identify missing values (NaN, null, or empty fields) and duplicate records. No missing values were found, and no duplicated image filenames were detected across the three dataset splits. A final sanity check confirmed that the number of images exactly matched the number of labels in each set.

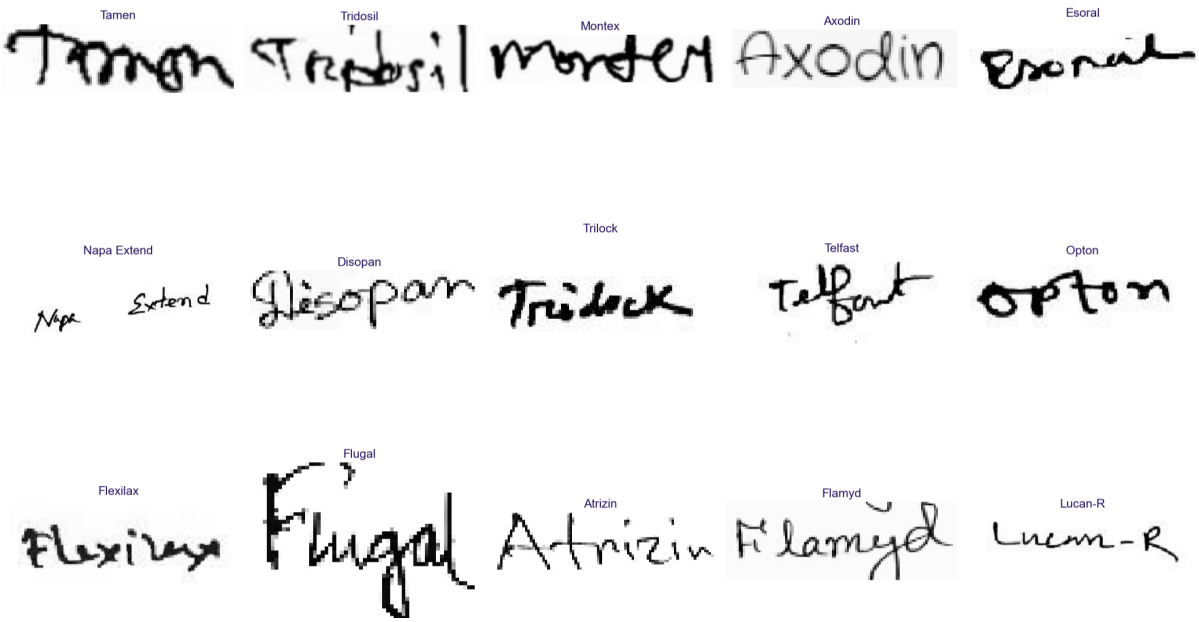
Sanity Check - no. of images vs no. of labels

```
# Check the number of images vs number of labels •••

Images: 3120 | Labels: 3120
Missing files = 0
```

Sampling Raw Images

Random sampling was performed using three batches of 15 images each to visually inspect image quality and verify label alignment. No cases of mismatched images and labels were identified.



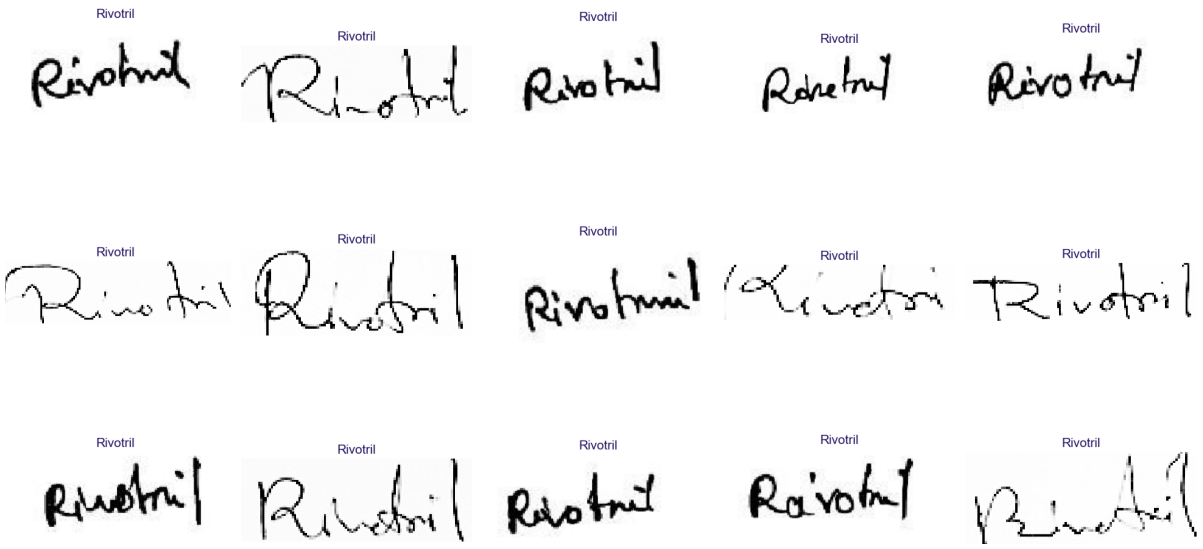
Sample raw images reveal natural variability in handwriting and image quality. Some samples are tightly cropped, while others show off-white backgrounds or grainy noise resulting from scan quality. These variations are representative of real-world prescription data.

To address this, standardised preprocessing is applied prior to modelling. Additionally, some samples exhibit slanted baselines, which is expected in rushed handwritten prescriptions. Cropped samples were intentionally retained, as similar issues are likely to occur in real-world usage scenarios.

Same class sampling

A focused review was conducted on 15 images from a single medication class to examine intra-class variation. An example using the medication *Rivotril* demonstrates substantial

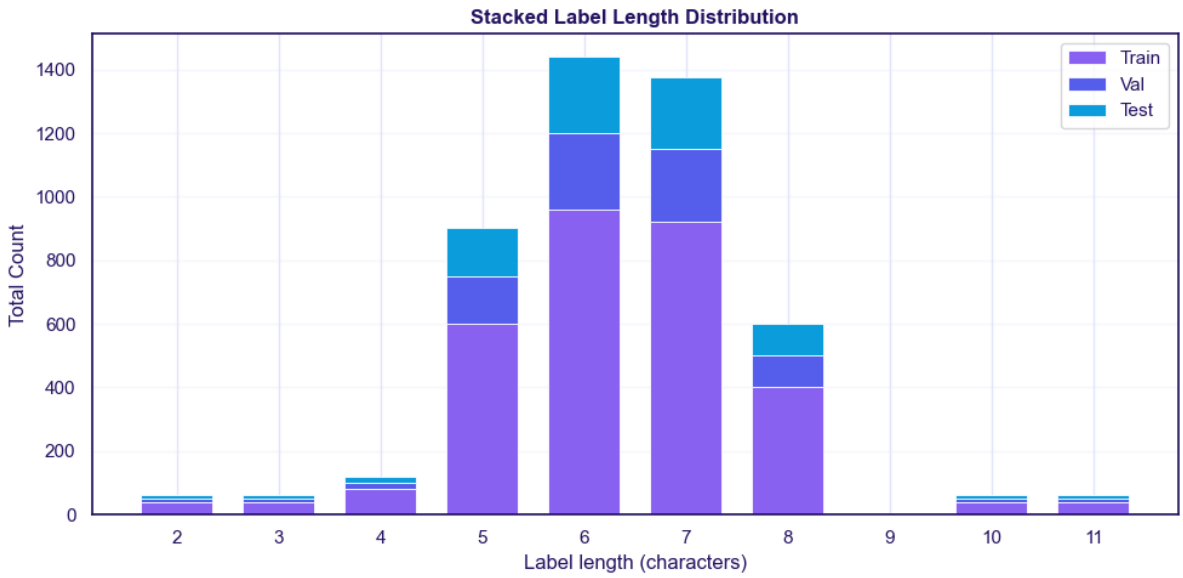
differences in handwriting style, stroke thickness, spacing, and image quality. These variations further support the suitability of the dataset for training a robust handwriting recognition model.



Total Character Length per Class Label

Analysis of medication name character lengths across the training, validation, and test sets shows a consistent and even distribution. The median label length is **6 characters**, with approximately **79.5%** of medication names falling between **5 and 7 characters**. The shortest labels contain **2 characters**, while the longest contain **11 characters**. This consistency confirms that label complexity is evenly distributed across all dataset splits.

Supporting tables and grouped distribution charts are shown below.



Label Length Distribution Table & Grouped Distribution % Table

	count	min	25%	median	75%	mean	max
Train	3120.0	2.0	6.0	6.0	7.0	6.33	11.0
Test	780.0	2.0	6.0	6.0	7.0	6.32	11.0
Val	780.0	2.0	6.0	6.0	7.0	6.33	11.0

	Train	Test	Val
label_len			
0–4	5.13	5.26	5.13
5–7	79.49	79.36	79.49
8–11	15.38	15.38	15.38

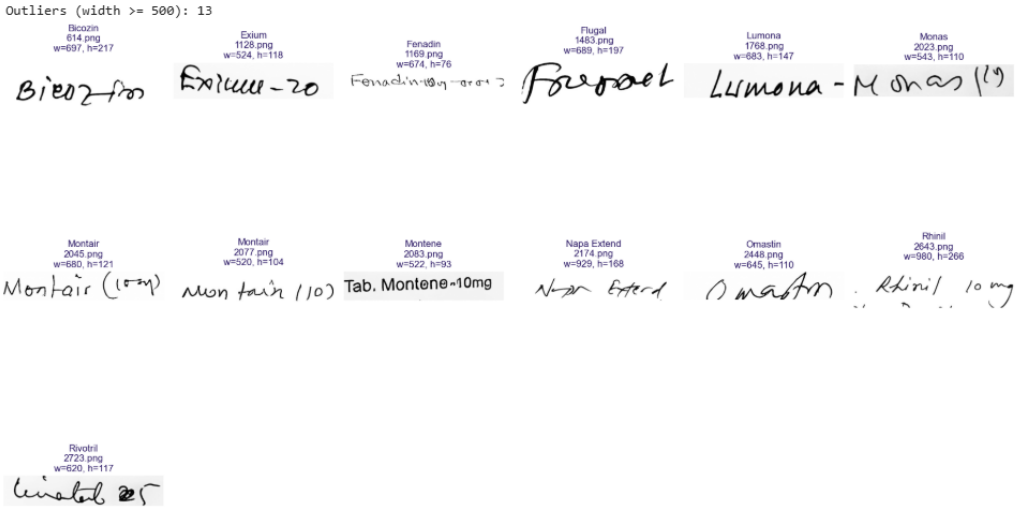
Examining Height and Width

A scatter plot was used to examine the distribution of image height and width across the raw dataset. The majority of images are clustered below **150 pixels in height** and **400 pixels in width**, indicating a relatively consistent size range for most handwritten medication samples.

The analysis also revealed a subset of outlier images located in the lower-right region of the scatter plot. These images have a relatively standard height but widths exceeding **500 pixels**. Further inspection showed that these wider samples typically include additional handwritten information beyond the medication name, most commonly dosage instructions.

Based on this observation, image preprocessing can be adjusted to standardise input dimensions. Specifically, image height can be reduced to a fixed target size, and a maximum width can be enforced. Images exceeding this width are likely to contain non-essential information, such as dosage details, which can be cropped without impacting the core task of medication name recognition.

Raw Images Greater than 500pixel Width



This approach helps reduce input variability, focuses the model on relevant features, and improves consistency across training samples.

Sampling Image Intensity - Pixel Intensity and Contrast

To assess overall image brightness and contrast consistency, a random sample of **100 training images** was analysed by computing the mean and standard deviation of pixel intensities for each image. All images were converted to grayscale, with pixel values ranging from **0 (black)** to **255 (white)**.

The **mean pixel intensity** represents the overall brightness of an image. Across the sampled images, the mean intensity was approximately **233**, indicating that most images are dominated by light backgrounds, with handwritten text contributing relatively small regions of darker pixels. This is consistent with scanned prescription images, where the background is typically white and handwriting occupies a limited portion of the image.

The **standard deviation of pixel intensities** reflects the level of contrast within each image, capturing how much pixel values vary between background and handwriting. The observed standard deviation of approximately **7.76** indicates low variation in contrast across the sampled images. This suggests that the majority of samples were scanned under similar conditions, resulting in relatively consistent contrast and minimal noise.

These findings support the use of light-weight intensity normalisation during preprocessing rather than aggressive contrast enhancement. Because brightness and contrast are generally consistent across samples

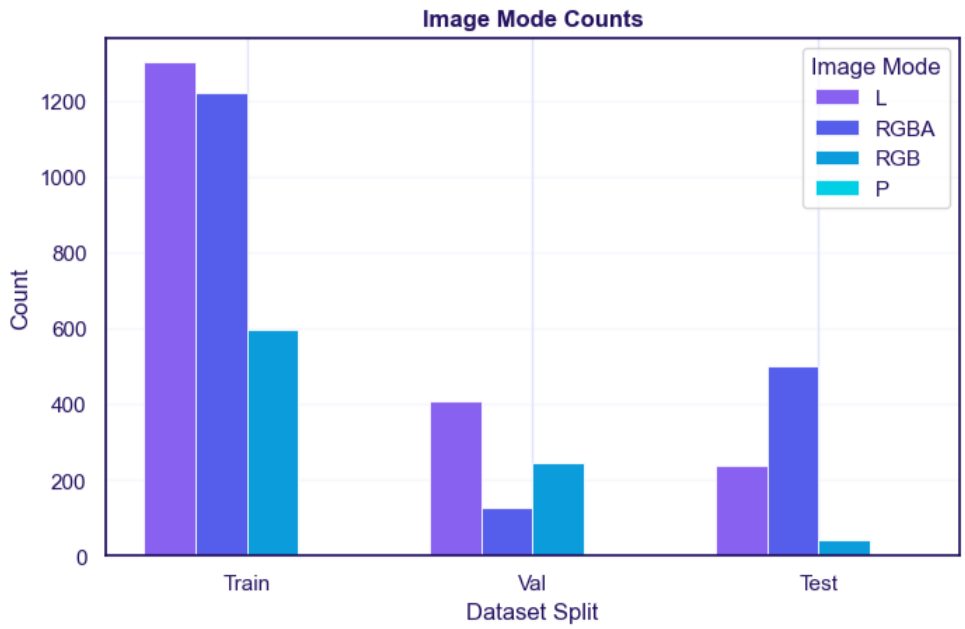
Image Mode

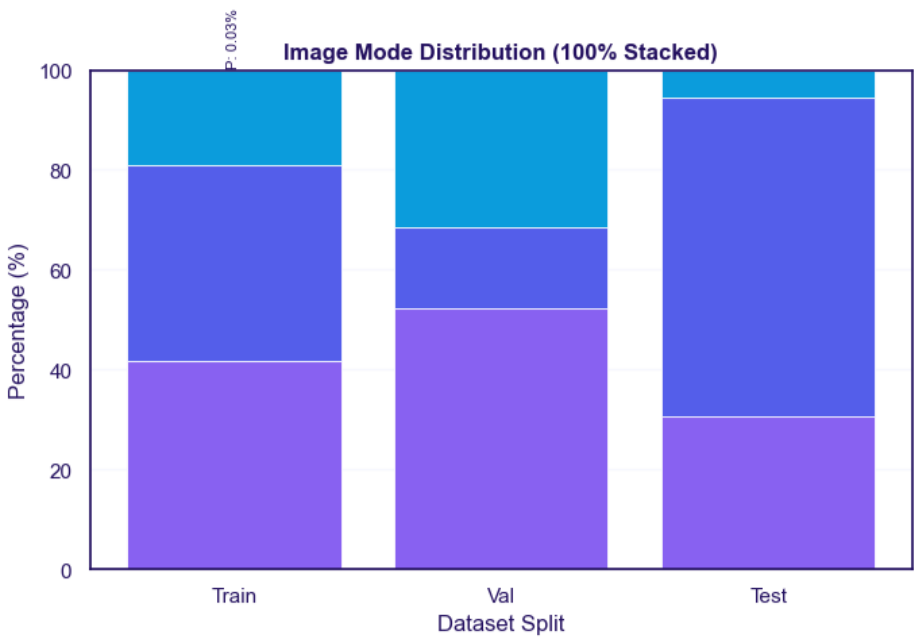
The Python Imaging Library (PIL) was used to inspect the **image mode** of the dataset, which indicates how pixel data is stored. This serves as a final data validation step to determine the level of preprocessing required before model training.

The images in the dataset fall into one of the following mode categories:

- **L** – Grayscale images with a single colour channel
- **RGB** – Colour images with three channels: Red, Green, and Blue
- **RGBA** – Colour images with four channels, including an alpha (transparency) channel
- **P** – Palette-based images, where pixel values reference an indexed colour palette

The distribution of Image Modes below:





Analysis of the image mode distribution shows that images are not stored uniformly across the dataset. As a result, all images are converted to **grayscale (L mode)** during preprocessing. This standardisation ensures consistent input dimensions and channel structure, reduces unnecessary colour information, and allows the convolutional neural network to focus on handwriting stroke patterns rather than colour-related artefacts.

Modelling

Image Preprocessing Pipeline

Before model training, all images are passed through a preprocessing pipeline designed to standardise input structure, reduce noise, and highlight handwriting features relevant to classification. Each step is informed by findings from the exploratory data analysis (EDA) and reflects real-world prescription variability.

Convert to Grayscale (Mode: "L")

Consistent with the earlier image mode analysis, all images are converted to grayscale (L mode) during preprocessing to ensure a consistent single-channel input.

Noise Reduction with Gaussian Blur

A light Gaussian blur is applied to reduce high-frequency noise introduced by scanning artefacts and paper texture. This decision is supported by EDA observations showing grainy backgrounds and scan-induced noise.

Contrast Enhancement

Contrast is enhanced using adaptive histogram equalisation to improve the separation between handwritten strokes and background. This ensures consistent stroke visibility across samples with varying scan quality without over-amplifying noise.

Mild Sharpening

A mild sharpening filter is applied to reinforce character edges and stroke boundaries following denoising and contrast enhancement. This helps preserve fine handwriting details and supports clearer feature extraction in the early convolutional layers.

Resize Image Height to 64 Pixels (Aspect Ratio Preserved)

Images are resized to a fixed height of 64 pixels while preserving the original aspect ratio. This choice is informed by EDA findings showing limited vertical variation in handwriting, allowing early layers to consistently capture stroke-level features while reducing redundant background space.

Width Normalisation to 384 Pixels (Padding or Cropping)

Image width is standardised to **384 pixels** by padding shorter samples and centre-cropping wider ones. EDA revealed that wider outliers often included dosage information

beyond the medication name, making cropping an effective way to remove non-essential content while retaining the core classification signal.

Pixel Normalisation

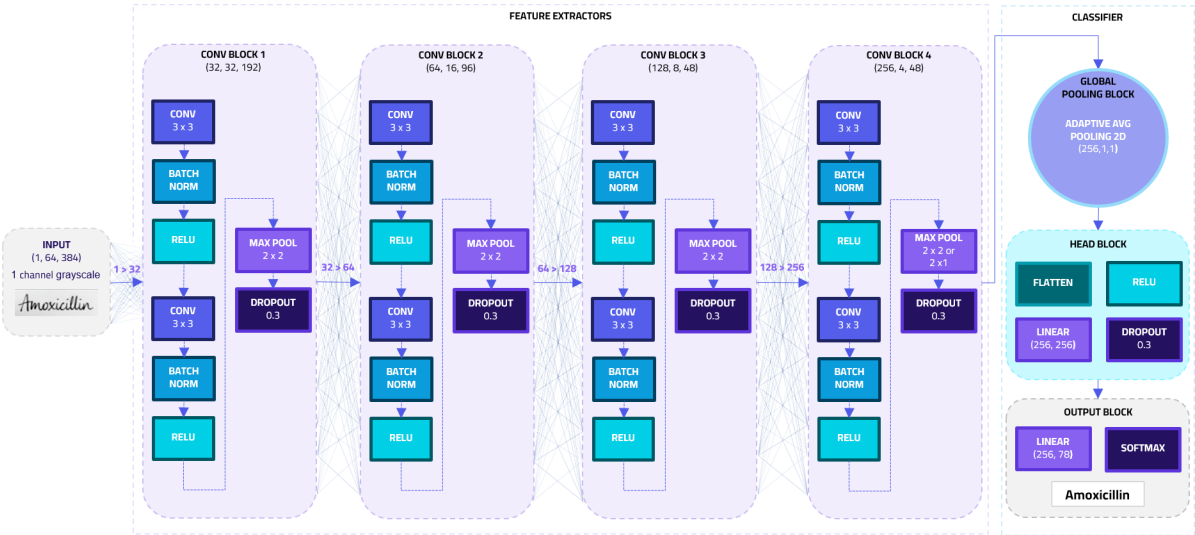
Pixel values are scaled to the range **[0, 1]** to stabilise optimisation during training. Given the observed consistency in brightness and contrast across samples, simple normalisation is sufficient to improve convergence without distorting handwriting features.

Lastly, the preprocessed images are exported as NumPy (.npy) arrays to enable fast, memory-efficient loading during model training. A new manifest file is generated to map each processed image to its label and original source.

Model Architecture

Given the project scope and timeline, this work focuses on optimising a single model architecture. A custom convolutional neural network (CNN) was designed for word-level handwritten medication name classification, leveraging fixed-size grayscale inputs (64 × 384 pixels) to enable efficient feature extraction, robustness to handwriting variation, and stable convergence on a moderately sized dataset.

Below is a graph representing the models architecture blocks and layers:



Convolutional Feature Extraction

The model is composed of a sequence of convolutional blocks, each containing stacked convolutional layers followed by batch normalisation, non-linear activation, and max pooling to downsize dimensions. These blocks progressively transform low-level stroke features (edges, curves, intersections) into higher-level representations that capture character shapes and word-level structure.

Stacking multiple convolutional layers within each block allows the receptive field to grow gradually, enabling the network to model both fine-grained handwriting details and broader spatial relationships across the width of the image. This design choice aligns with earlier EDA findings showing left-to-right word structure and limited vertical variation in handwriting.

Normalisation and Regularisation

Batch normalisation is applied throughout the network to stabilise training and reduce sensitivity to scale differences in feature activations. Dropout is incorporated in deeper layers to mitigate overfitting, which is particularly important given the relatively small number of samples per class and the high visual similarity between certain medication names.

Global Average Pooling and Classification Head

Following convolutional feature extraction, spatial information is aggregated using pooling operations that reduce dimensionality while preserving features. The resulting feature maps are passed to a fully connected classification head, which outputs class probabilities across the **78 medication name classes**.

The final layer uses a softmax activation, enabling probabilistic interpretation of predictions and supporting both **Top-1** and **Top-5** evaluation metrics.

Training Strategy & Optimisation

The training strategy for this project was designed to maximise classification performance while maintaining stability, generalisation, and reproducibility. Given the moderate dataset size and the high visual similarity between certain medication names, particular emphasis was placed on preventing overfitting and ensuring balanced performance across all classes. The trained model was then evaluated on validation and test data to assess convergence, robustness, and real-world generalisation.

Optimisation and Loss Function

Training used categorical cross-entropy loss with an adaptive optimiser to efficiently converge on a multi-class classification objective. Label smoothing was applied to reduce overconfident predictions and improve separation between visually similar medication names.

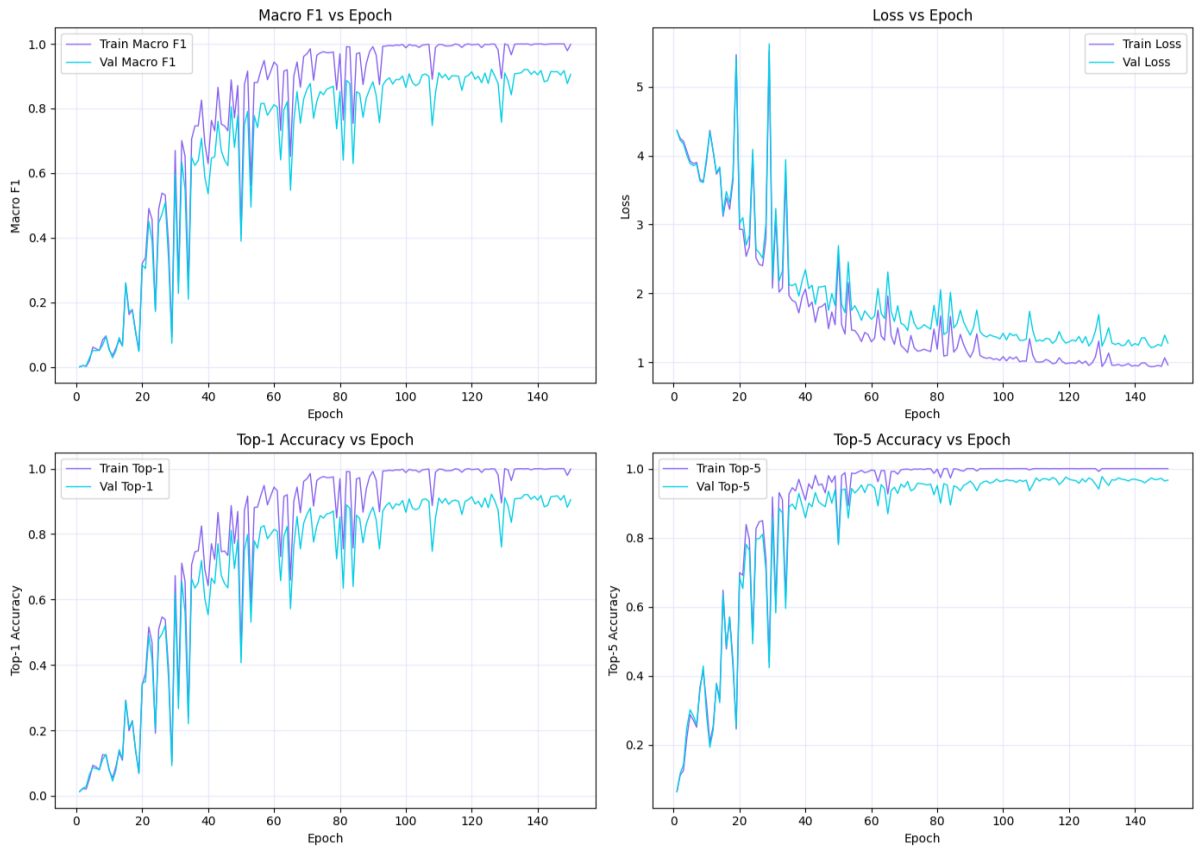
Regularisation and Overfitting Control

Dropout, batch normalisation, and training-only data augmentation were used to improve generalisation and stabilise learning under handwriting variability. Early stopping and checkpointing ensured the model retained the best validation performance without overfitting.

Overall Model Performance

Model performance was evaluated using **Top-1 accuracy** as the primary metric, with **Top-5 accuracy**, **Macro F1-score**, and **loss** used as supporting measures. The charts below present the training and validation curves across epochs, providing insight into convergence behaviour, generalisation, and performance stability.

Training and validation performance across epochs (Macro F1, Top-1, Top-5, Loss)



Training was run for approximately **150 epochs** to allow full observation of convergence behaviour and ensure that performance improvements had stabilised. As shown, validation Top-1 accuracy increases steadily before plateauing at approximately **92%** around the best validation epoch, while validation Top-5 accuracy converges earlier and remains high at approximately **97%**, indicating that the correct medication name is typically present among a small set of candidate predictions when Top-1 classification fails.

Class-Level Performance and Robustness

Macro F1-score curves show stable convergence with a small gap between training and validation performance, indicating balanced classification across all 78 medication classes. This suggests the model generalises well across handwriting styles without bias toward visually distinct labels.

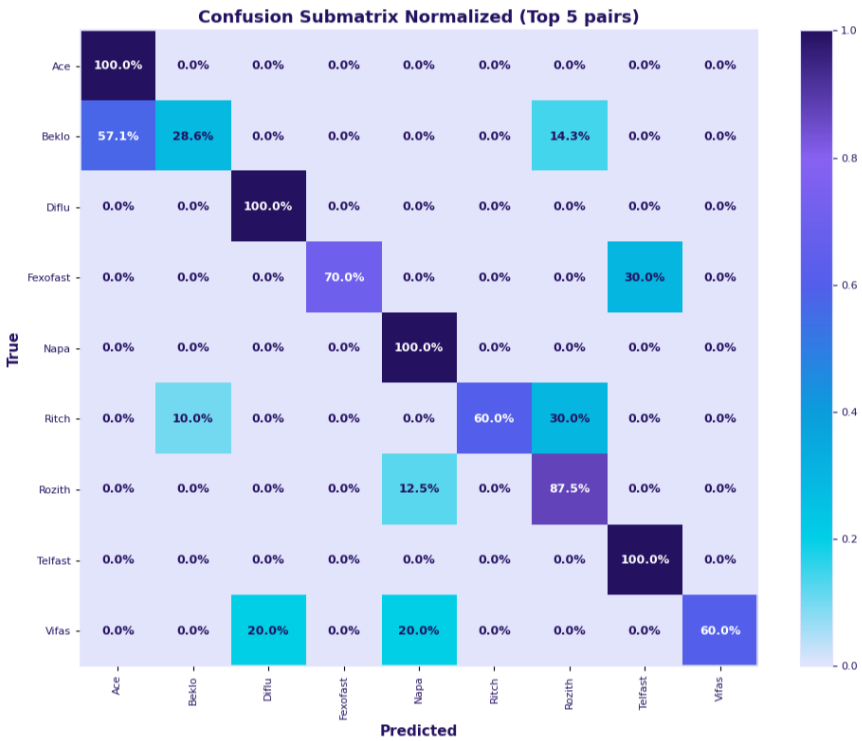
Real-World Test: Model Inference on Unseen Handwriting

rank	prediction	probability_%
1	Flexilax	17.89
2	Provair	8.82
3	Trilock	3.99
4	Diflu	3.59
5	Dinafex	3.42

This is further supported by inference on an unseen, real-world handwritten sample, where the correct medication name appears first among the model's top-5 predictions.

Error Analysis and Misclassifications

Loss curves show smooth convergence with no significant divergence between training and validation loss, indicating controlled overfitting and stable generalisation. Remaining errors are primarily associated with visually similar medication names or highly ambiguous handwriting samples, rather than systematic model failure.



The above presents a normalised confusion submatrix for the most frequently confused medication pairs, showing that misclassifications are concentrated among a small subset of visually similar classes while most classes retain strong diagonal dominance. This supports the Macro F1 results and indicates that classification errors are structured and predictable rather than random.

Top-5 Misclassified

The Top-5 misclassified sample further illustrates these patterns through examples of predictions on unseen test samples. In these cases, the correct medication name frequently appears within the Top-5 predictions, highlighting the model's ability to narrow ambiguity effectively and reinforcing its suitability for assisted decision-making workflows.

Outcomes

On the held-out test set, The final model achieved a **Top-1 accuracy of 82.2%**, **Top-5 accuracy of 92.9%**, and **Macro F1-score of 81.3%**, demonstrating consistent performance across all 78 medication classes. These results indicate the model generalises effectively to unseen handwriting and is suitable for semi-automated prescription workflows where high-confidence predictions can be actioned and ambiguous cases reviewed.

Data Answer

Building on the outcomes above, while the achieved results demonstrate strong and reliable performance, there remains clear opportunity for further improvement. Future work could focus on optimising the existing model through additional tuning and augmentation, expanding the dataset to include real-world Australian prescription samples, and extending the task to recognise dosage and multi-line text. In addition, exploring sequence-aware or transformer-based architectures may further improve performance on highly ambiguous handwriting, particularly in cases where character ordering and context play a greater role.

Business Answer and Response to Stakeholders

The results of this project indicate that a digital method for submitting handwritten prescriptions online is feasible as a proof of concept, supporting the introduction of a third submission pathway alongside mailed scripts and eScripts. While not intended as a fully autonomous solution, the model demonstrates sufficient reliability to show potential for increasing online adoption and reducing manual review effort through assisted prescription validation, subject to regulatory approval.

End-to-End Solution / Implementation

The observed performance supports the feasibility of using the model in a customer-facing application where high-confidence predictions can be processed automatically and lower-confidence cases escalated for review.

The proposed end-to-end solution enables patients to submit handwritten prescriptions digitally, where images are preprocessed, classified by a handwriting recognition model, and integrated into pharmacy workflows to support fast, semi-automated validation with human oversight.

References

- eScripts issued in 2024 https://www1.racgp.org.au/newsgp/professional/funding-boost-will-see-eprescribing-expand-into-ho#:~:text=expand%20into%20hospitals-,newsGP%20writers,prescriptions%20have%20been%20electronic%20prescriptions
- Prescriptions issued 2022-2023 https://www.aihw.gov.au/reports-data/health-welfare-services/medicines/overview#:~:text=Medicines%20in%20the%20health%20system&text=In%202022%E2%80%9323%2C%20335.8%20million,3.9%25%20on%202021%E2%80%9322.
- IBISWorld Pharmacy industry https://www.ibisworld.com/australia/industry/pharmaceutical-product-manufacturing/188/
- IBISWorld – Pharmacy industry Market share https://www.ibisworld.com/australia/industry/pharmacies/1878/
- VGG Explained https://www.geeksforgeeks.org/computer-vision/vgg-net-architecture-explained/
- EDA for handwritten image recognition: https://www.kaggle.com/code/chaimaourgani/quick-eda-on-the-rimes-handwritten-dataset#EDA---Dataset-Overview
- Handwritten image recognition with NNs:| https://www.geeksforgeeks.org/machine-learning/handwritten-digit-recognition-using-neural-network/
- Handwritten recognition EDA & Modelling: https://www.kaggle.com/code/vineeth1999/handwritten-recognition-eda-and-modelling/notebook
- CNN Explainer https://poloclub.github.io/cnn-explainer/#article-convolution
- Image Kernels explained https://setosa.io/ev/image-kernels/
- VGG Explained https://www.geeksforgeeks.org/computer-vision/vgg-net-architecture-explained/
- AdaptiveAvgPooling2D Documentation https://docs.pytorch.org/docs/stable/generated/torch.nn.AdaptiveAvgPool2d.html
- Pretrained Model - tuning or transfer Learning https://keras.io/guides/transfer_learning/
- OvO vs OvR pros and cons - Kaggle https://www.kaggle.com/discussions/general/544292
- OvO vs OvR pros and cons – machine learning https://machinelearningmastery.com/one-vs-rest-and-one-vs-one-for-multi-class-classification/